Md Rezwan Bin Mizan

Department of Mathematics, University of Houston

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Academic Appointments

- **Instructor of Record** *Aug 2025 – Present*
  University of Houston
- **Research Assistant** *Jun 2024 – Present*
  University of Houston
- **Graduate Teaching Assistant** *Aug 2021 – May 2024*
  University of Houston
- **Lecturer** *Jan 2018 – Jul 2021*
  American International University–Bangladesh (AIUB)

Education

- **Ph.D. in Mathematics** *Aug 2021 – Expected May 2026*
  University of Houston, Texas, USA
  - Dissertation (in progress): Tensor-Based Reduced-Order Modeling for Parametric Nonlinear PDEs
  - Advisor: Dr. Maxim Olshanskii Co-Advisor: Dr. Ilya Timofeyev
- **M.S. in Applied Mathematics** *Dec 2025*
  University of Houston, Texas, USA
- **M.S. in Applied Mathematics** *2016 – 2017*
  University of Dhaka, Bangladesh
  - Thesis: Spectral Relaxation Approach on Convective Heat Transfer of Magnetic and Non-Magnetic Particles over a Stretching Cylinder with Induced Magnetic Enhancement.
- **B.S. in Mathematics** *2010 – 2015*
  University of Dhaka, Bangladesh
  - Project: A Study of Differential Equations in Population Biology.

Research Interests

- Reduced Order Modeling for nonlinear and parametric partial differential equations
- Tensor-based methods and machine learning-enhanced ROMs (DL-ROM, hybrid approaches)
- Computational fluid dynamics, including non-Newtonian and viscoplastic flows (Herschel–Bulkley models)
- Large-scale scientific computing and efficient simulation of complex systems

Publications

Published

- **A Parametric Tensor ROM for the Shallow Water Dam Break Problem.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  Computers & Fluids, 2026. DOI: 10.1016/j.compfluid.2026.107005.
- **Mathematical and Buongiorno Nanofluid Modelling to Predict the Thermal and Solutal Performance of Magneto-Bioconvective Casson Nanofluid Flowing in a Rotatory Disk.**
  Sulyman O. Salawu, MD Shamshuddin, **Md Rezwan Bin Mizan**, and Sajjadul Bari.
  Physica Scripta, 2024. DOI: 10.1088/1402-4896/ad8e1a.
- **Computation of Ferromagnetic/Nonmagnetic Nanofluid Flow Over a Stretching Cylinder with Induction and Curvature Effects.**
  Md Rezwan Bin Mizan, Mohammad Ferdows, MD Shamshuddin, et al.
  Heat Transfer, 2021. DOI: 10.1002/htj.22122.

Under Review

- **Parametric Reduced Order Models for the Generalized Kuramoto–Sivashinsky Equations.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  Under review at Applied Mathematics and Computation, 2025.
  Preprint: arXiv:2502.02718.
- **Tensorial Reduced-Order Models for Parametric Coupled Reaction-Diffusion Systems: Application to Brain Tumor Growth Modeling.**
  Asikul Islam, **Md Rezwan Bin Mizan**, Maxim Olshanskii, and Andreas Mang.
  Under review at Computers and Mathematics with Applications, 2026.
  Preprint: arXiv:2603.14101.

In Preparation

- **Reduced Order Modeling of Parametrized Visco-Plastic Shallow Fluid Flow.**
  Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
  In preparation.

Teaching Experience

- **Instructor of Record** *Aug 2025 – Present*
  University of Houston
  - Design and deliver a computation-driven linear algebra course integrating theoretical concepts with MATLAB-based applications.
  - Develop assignments, exams, and coding projects emphasizing conceptual understanding and problem solving.
  - Mentor students through office hours and support diverse learning needs.

  Fall 2025: MATH 2131 – Linear Algebra with MATLAB
  Spring 2026: MATH 2131 – Linear Algebra with MATLAB

- **Graduate Teaching Assistant** *Aug 2021 – May 2024*
  University of Houston
  - Led Calculus I–III lab sessions and supported instruction across multiple undergraduate courses.
  - Provided tutoring at CASA and assisted students with diverse mathematical backgrounds.
  - Evaluated coursework in calculus, discrete mathematics, probability, and linear algebra.

  Fall 2021: MATH 2413 – Calculus I
  Spring 2022: MATH 2413 – Calculus I; MATH 3336 – Discrete Mathematics
  Summer 2022: MATH 3338 – Probability *(grader/tutor)*
  Fall 2022: MATH 3338 – Probability *(grader/tutor)*
  Spring 2023: MATH 2414 – Calculus II
  Summer 2023: MATH 4380 – Survey of Mathematics *(grader/tutor)*
  Fall 2023: MATH 4380 – Survey of Mathematics *(grader/tutor)*
  Spring 2024: MATH 2305 – Discrete Mathematics
  Fall 2024: MATH 2318 – Linear Algebra
  Summer 2025: MATH 2415 – Calculus III

- **Lecturer** *Jan 2018 – Jul 2021*
  American International University–Bangladesh (AIUB)
  - Taught undergraduate courses including Calculus, Differential Equations, Numerical Methods, and Complex Variables.
  - Designed course materials and assessments aligned with learning objectives.
  - Transitioned courses to online delivery during COVID-19 and mentored students on academic progress.

Conference and Seminar Talks

- **A Parametric Tensor ROM for the Shallow Water Dam Break Problem**
  Minisymposium, SIAM TX–LA Conference, University of Texas at Austin, Sep 2025.
- **Tensorial Reduced Order Modeling of Viscoplastic Shallow-Water Dam-Break Flows**
  Scientific Computing Seminar, University of Houston, Sep 2025.
- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation**
  Graduate Student Presentation, University of Houston, Apr 2024.

Poster Presentations

- **POD–ROMs for the Generalized Kuramoto–Sivashinsky Equation**
  SIAM TX–LA Section Meeting, Baylor University, Oct 2024.
- **Reduced Order Modeling of Kuramoto–Sivashinsky Equation using Proper Orthogonal Decomposition**
  Mathematical Opportunities in Digital Twins (Math-DT), George Mason University, Dec 2023.

Leadership & Service

- **Treasurer** *2023 – Present*
  AMS Student Chapter, University of Houston
  - Manage chapter finances and support organization of academic and student-focused events.
- **Public Relations & Publicity Secretary** *2025 – 2026*
  Bangladesh Graduate Student Association, University of Houston
  - Coordinate communications and promote events to enhance student engagement and community outreach.

Technical Skills

- **Programming**: Python, MATLAB, FORTRAN, Mathematica
- **Scientific Computing**: Numerical methods for PDEs, finite volume methods, reduced order modeling (ROM)

Languages

- English (fluent), Bengali (native)

Professional Training

- CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning, University of Houston, 2025
- TAMIDS Scientific Machine Learning (SciML) Summer School, Texas A&M University, 2025
- Workshop on Presentation, Goal Setting, and Professional Development, Life Carnival Foundation
- Faculty Development Webinar on Outcome-Based Education (OBE), AIUB

Honors & Fellowships

- **NSF-funded Research Assistantship (DMS-2309197, DMS-2408978)** *2024 – Present*
  University of Houston
- **Graduate Tuition Fellowship** *2021 – Present*
  University of Houston
- **Teaching Assistantship** *2021 – 2024*
  University of Houston
- **Bangladesh Sweden Trust Fund Award** *2021*
  Ministry of Finance, Bangladesh
- **National Science and Technology Fellowship** *2016 – 2017*
  Government of Bangladesh

Memberships

- **Member** *2022 – Present*
  American Mathematical Society (AMS)
- **Member** *2023 – Present*
  Society for Industrial and Applied Mathematics (SIAM)